Image artifacts, region-of-interest masking and shortcut learning in medical image classification: a review of work published from 2021 to 2026

MSc thesis *Where the Shortcut Sits* — literature review

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1 Introduction

A classifier takes a shortcut when it relies on something that predicts the label in the training data without being evidence of the disease (Geirhos et al., 2020). In medical imaging such signals are rarely random. Clinicians measure suspicious nodules, mark lesions they intend to excise and drain collapsed lungs, so calipers, skin markings and chest tubes appear more often in the images of sick patients. A common response is to segment the lesion or organ and let the model see only that region of interest (ROI). The thesis asks when this works, and the answer it gives is that it depends on where the artifact lies: masking removes an artifact outside the ROI but keeps one inside it, and the model then relies on it more.

This review covers peer-reviewed work from 2021 to October 2026 that bears on that question. We kept to journals and main-track conferences with an established reputation in medical imaging and machine learning, and left out preprints and workshop papers; a few older papers are cited where later work builds on them. The review is organised in seven parts: the evidence that medical classifiers take shortcuts, the artifacts documented in each modality, what is known about restricting a model to the ROI, how shortcuts are detected, how they are removed, why models prefer them, and how claims of this kind should be evaluated. A final section states where the thesis stands relative to this work.

2 Evidence that medical classifiers take shortcuts

The COVID-19 literature made the problem hard to ignore. DeGrave et al. (2021) showed that radiographic COVID-19 detectors relied on laterality markers, patient positioning and other differences between the source datasets, and that their accuracy fell on data from other hospitals; they used saliency maps and generative counterfactuals to show what the models attended to. A systematic review of 62 published COVID-19 imaging models found none fit for clinical use, mainly because of bias in small datasets and the variability of large ones assembled from many sources (Roberts et al., 2021). The most direct demonstration came from Maguolo and Nanni (2021), who still classified COVID-19 radiographs after blacking out the lungs. If the label can be predicted with the anatomy removed, the information must be somewhere else in the image. This is exactly the reasoning behind ROI masking, and it is worth noting that it says nothing about shortcuts that sit *inside* the anatomy.

A second line of work concerns patient attributes. Deep models predict self-reported race from medical images, even when the images are heavily degraded (Gichoya et al., 2022), and chest-radiograph classifiers underdiagnose under-served groups, most strongly at intersections such as Hispanic female patients (Seyyed-Kalantari et al., 2021). Glocker et al. (2023) found that

disease models encode protected characteristics in their features, and Yang et al. (2024) showed that the models encoding demographics most strongly have the largest fairness gaps, and that a model made fair in one hospital is not necessarily fair in another. Encoding an attribute is not the same as using it, and Brown et al. (2023) proposed shortcut testing to distinguish the two by varying how strongly the attribute is encoded. Jones et al. (2024) argue that dataset biases which look alike in the data can arise from different causal mechanisms (prevalence, presentation and annotation) and need different remedies. We find this framing useful: the artifacts in the thesis are produced by clinical actions taken on suspicion, so they are part of how the data are generated and cannot be cleaned away as noise.

Several groups study shortcuts under controlled conditions. Stanley et al. (2024) built synthetic brain images with known disease and bias effects, placed the bias either near the disease region or far from it (both outside the region itself), and found that this choice changed the performance gaps and how well three mitigation strategies worked (reweighing worked best); Sun et al. (2023) added a tag, hyperintensities or an obstruction to chest radiographs and found that explanation methods differ widely in whether they reveal the injected signal. Stanley et al. are therefore the closest precedent for the thesis: they show that where a bias lies matters. They do not study ROI masking, and their bias is synthetic. The thesis varies an artifact's position relative to the ROI that masking keeps, with pasted and real artifacts.

3 Artifacts documented in practice

Dermoscopy has the richest record. Ruler scale bars changed the melanoma predictions of a market-approved classifier (Winkler et al., 2021), as surgical skin markings had done in an earlier study (Winkler et al., 2019). Explanation-based audits of ISIC models uncovered colour calibration patches, rulers and other artifacts (Anders et al., 2022; Pahde et al., 2023). Hair that covers the lesion degrades classifiers (Wang et al., 2024), and for dark corners, adding synthetic ones during training helped more than removing the real ones (Pewton et al., 2024). The ISIC collections also contain duplicate images across releases, which can leak between training and test sets (Cassidy et al., 2022). Read together, these studies show that the artifacts sit in different places: patches and dark corners lie mostly outside the lesion, rulers near the image edge, markings around the lesion, and hair crosses it. The studies report the effect of an artifact being present; none of those we reviewed asks whether its position matters.

In ultrasound the evidence is thinner. Lin et al. (2024) showed that calipers and on-screen text act as shortcuts in fetal ultrasound, even for segmentation. For thyroid ultrasound, one validation study considered calipers as a possible confound and judged that differences in their appearance between scanners did not explain its results (Weng et al., 2023); we found no study that measures calipers as a classification shortcut in thyroid or ovarian ultrasound, although calipers are placed on the nodule border by design. In chest radiography, pneumothorax classifiers respond to the chest tube that treats the condition. Rueckel et al. (2021) reduced this confounding by training with in-image annotations, and Jiménez-Sánchez et al. (2023) showed that a CheXpert-trained model loses AUROC on NIH images without drains (81.1% with drains, 65.5% without); the drain annotations were later released as NEATX (Cheplygina et al., 2025). A drain projects over the lung field, so a lung mask cannot remove it. For capsule endoscopy we found no study of bubbles or debris as a shortcut, even though debris and lesions often occupy the same pixels.

4 Does restricting the model to the ROI help?

In natural images, classifiers can recognise objects from the background alone, and more accurate models tend to rely on it less (Xiao et al., 2021). The same holds for medical images. In a careful

masking study on chest radiographs and fundus photographs, models trained on full images performed better on images with the ROI removed than on images containing only the ROI (Sourget et al., 2026). Training on segmented dermoscopy images improved balanced accuracy for classifiers trained on HAM10000 but not for those trained on ISIC (Maron et al., 2021), a mixed result that is easy to overlook. ISNet takes a softer route and penalises relevance outside the segmentation mask during training, which made chest-radiograph classifiers more robust to background bias (Bassi et al., 2024); its evaluation already uses matched, reversed and artifact-free tests, with an artificial mark that always lies outside the lungs.

All of these studies assume that the shortcut lies in the background. The clearest result pointing the other way comes from natural images: when several shortcuts are present, removing one can make the model rely more on the others (Li et al., 2023). ROI masking is a mitigation of exactly this kind. If an artifact remains inside the ROI, the same reasoning predicts that masking will increase reliance on it. To our knowledge this has not been tested in medical images, and it is the starting point of the thesis.

5 Detecting shortcuts

Explanation methods are the most common tool. Spectral relevance analysis clusters heatmaps to reveal Clever-Hans strategies (Anders et al., 2022), and Reveal-to-Revise turns detection, correction and re-evaluation into a cycle (Pahde et al., 2023). The limits are well documented: saliency maps localise chest pathology worse than radiologists (Saporta et al., 2022), and their ability to reveal a known confounder varies from method to method (Sun et al., 2023). In our view there is a further limit for in-ROI artifacts: when calipers sit on a nodule, a heatmap that highlights the nodule cannot tell whether the model looks at the tissue or at the calipers.

Counterfactual methods avoid this by changing the image. Diffusion-based counterfactuals add or remove a suspected shortcut and measure the change in prediction (Weng et al., 2024), and RadEdit edits chest radiographs under masks to stress-test classifiers (Pérez-García et al., 2024). The thesis uses the same idea in its simplest form, by pasting one real artifact inside or outside the ROI of the same images.

6 Removing shortcuts

With group labels, retraining only the last layer on balanced data recovers much of the robustness lost to spurious correlations (Kirichenko et al., 2023), because standard training already learns the core features even when the classifier does not use them (Izmailov et al., 2022). Without group labels, the errors of a first model can be up-weighted (Liu et al., 2021), or the last layer reweighted automatically (Qiu et al., 2023); variance-of-risk penalties are another option (Krueger et al., 2021). The benchmarks are sobering. Across 20 algorithms and 12 datasets, methods improved subgroup robustness only for some kinds of shift (Yang et al., 2023), and in medical imaging, bias-mitigation methods did not improve fairness significantly over standard training, while the choice of model selection mattered more (Zong et al., 2023).

A more principled line starts from causal reasoning. Counterfactual invariance requires that the prediction does not change when only the nuisance changes; when the label causes the image, as in diagnosis, this implies independence of the prediction and the nuisance given the label (Veitch et al., 2021). Makar et al. (2022) enforce such conditional independences with auxiliary labels available at training time, and NuRD trains under a distribution in which the nuisance–label relation is broken (Puli et al., 2022). Kaur et al. (2023) show that the right constraint depends on how the data were generated, so no single constraint suits every shift. Linear concept erasure removes a concept from the features in closed form (Belrose et al., 2023), at the cost of

also removing label signal when the two are correlated; SPLINCE was designed to keep that signal (Holstege et al., 2025). Generative data are a different route: diffusion-generated images improved fairness out of distribution in histopathology, chest radiography and dermatology (Ktena et al., 2024).

The remedy the thesis recommends, a class-conditional mean constraint on the masked features, is a first-moment, linear case of the conditional independence of Veitch et al. (2021) and Makar et al. (2022). What the thesis adds is the reason for applying it after masking, and a registered test of whether any remedy beats masking across all cohorts. Given the benchmark results above, it is not surprising that none does.

7 Why models prefer shortcuts

Theory explains why a simple feature wins even when a reliable one is available. Nagarajan et al. (2021) show that geometric and statistical skews push standard training toward spurious features even when the invariant feature predicts the label perfectly. Gradient descent with cross-entropy can starve less salient features (Pezeshki et al., 2021), and its implicit preference for large margins produces shortcut reliance even when the shortcut adds no information (Puli et al., 2023). Hermann et al. (2024) separate a feature's *predictivity* from its *availability*, that is, how easily it can be extracted, and measure shortcut bias as reliance on the more available but less predictive feature. For frozen pretrained features, as in most of the thesis, linear probing can generalise better out of distribution than full fine-tuning (Kumar et al., 2022). The thesis's linear-Gaussian account fits this picture: masking removes competing evidence outside the ROI and so changes the relative predictivity of what is left. The difference is that the account was used to make quantitative predictions that were registered before the results existed.

8 How such claims should be evaluated

Varoquaux and Cheplygina (2022) review methodological failures in medical-imaging machine learning, from biased data to publication incentives, and Kapoor and Narayanan (2023) found data leakage, including non-independent splits, in at least 294 papers across 17 fields. Results also vary with data sampling, initialisation and hyperparameters, and accounting for several of these sources at once gives better estimates of that variation (Bouthillier et al., 2021). In MICCAI papers, confidence intervals are rarely reported, and their typical width is larger than the gap between the first- and second-ranked methods (Christodoulou et al., 2024). Overall metrics also hide failures in clinically important subgroups (Yang et al., 2024), a problem first described as hidden stratification (Oakden-Rayner et al., 2020). These papers shaped the evaluation in the thesis: splits grouped by patient or lesion, a bootstrap that resamples training seeds and test images together, registration before results, and evaluation on the cases that contradict the shortcut as well as on the whole test set. The correlated, reversed and clean tests are borrowed as well. Colored MNIST flips a colour cue between training and test (Arjovsky et al., 2019); Puli et al. (2022) train chest-radiograph models at 90/10 agreement between hospital source and label and test at 10/90; Bassi et al. (2024) and Corbetta et al. (2026) evaluate on reversed and clean sets; and a 2020 workshop study reversed seven dermoscopy artifact associations and evaluated lesion-exterior masking. What the thesis adds is not the test but the variable it is crossed with: the artifact's position relative to the ROI that masking keeps.

9 Where the thesis stands

The literature establishes four things firmly. Medical classifiers take shortcuts, including ones produced by clinical practice; the artifacts involved are well documented in dermoscopy and chest radiography; image backgrounds carry signal, and restricting a model to the ROI sometimes helps; and no mitigation method is reliably better than standard training across settings. Table 1 lists what remained open and how the thesis addresses it.

Table 1: Open questions in the reviewed work and the corresponding contribution of the thesis.

Open question	Closest prior work	Thesis
Does the artifact's location relative to the ROI decide whether masking works?	Sourget et al. (2026); Maron et al. (2021); Bassi et al. (2024): masking studies that assume a background shortcut; Boland et al. (2024): fixed vs random shortcut position (no ROI); Stanley et al. (2024): a synthetic bias's distance from the disease region matters (no masking)	Location varied directly: controlled sweeps, traps built from real artifacts, and a transplant of the same artifact across the ROI boundary
Is the test design new?	Arjovsky et al. (2019); Puli et al. (2022); Bassi et al. (2024); Corbetta et al. (2026): correlated, reversed and clean tests	No; the thesis borrows it and crosses it with the artifact's position relative to the ROI and with masking
Can masking increase reliance on an artifact?	Li et al. (2023): removing one shortcut amplifies others (natural images)	Reliance on an in-ROI artifact grows after masking, in four cohorts and three modalities
Can the harm be predicted?	Hermann et al. (2024); Puli et al. (2023): explanatory theories	A two-parameter account with predictions registered before the results
What should be used instead of masking?	Veitch et al. (2021); Makar et al. (2022); Zong et al. (2023)	A registered test against masking; a class-conditional constraint comes closest, none dominates
Calipers and debris as classification shortcuts	Lin et al. (2024) (segmentation only)	Thyroid and ovarian ultrasound and capsule endoscopy alongside dermoscopy

Some of this is not new, and the thesis should say so. The reversed test is established (Arjovsky et al., 2019; Puli et al., 2022; Bassi et al., 2024; Corbetta et al., 2026). That a mask cannot remove what it keeps is self-evident; what was missing is evidence that masking makes reliance on the kept artifact grow, how large the effect is on real artifacts, and how to predict it. The recommended constraint is a special case of known conditional-invariance methods, and the group-robustness methods are used as baselines rather than proposed. The closest conceptual precedent is the whac-a-mole result of Li et al. (2023); the thesis supplies the location-specific mechanism and the medical evidence.

Use of AI tools

This review was prepared with the assistance of an AI system (Claude, Anthropic), which was used to search the literature, check bibliographic details against publisher and proceedings pages, and draft the text. The choice of scope, the interpretation of the studies and any remaining errors

are the author's responsibility.

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